

Module 3

Fundamentals of frequency modulation

Frequency modulation: The carrier frequency is varied in accordance with the message signal by keeping amplitude and phase constant.

As the amplitude of the information signal varies, the carrier frequency shifts proportionately. As the modulating signal amplitude increases, the carrier frequency increases. If the amplitude of the modulating signal decreases, the carrier frequency decreases.

The amount of change in carrier frequency produced by the modulating signal is known as the *frequency deviation* f_d . Maximum frequency deviation occurs at the maximum amplitude of the modulating signal.

The frequency of the modulating signal determines the frequency deviation rate, or how many times per second the carrier frequency deviates above and below its center frequency. If the modulating signal is a 500-Hz sine wave, the carrier frequency shifts above and below the center frequency 500 times per second.

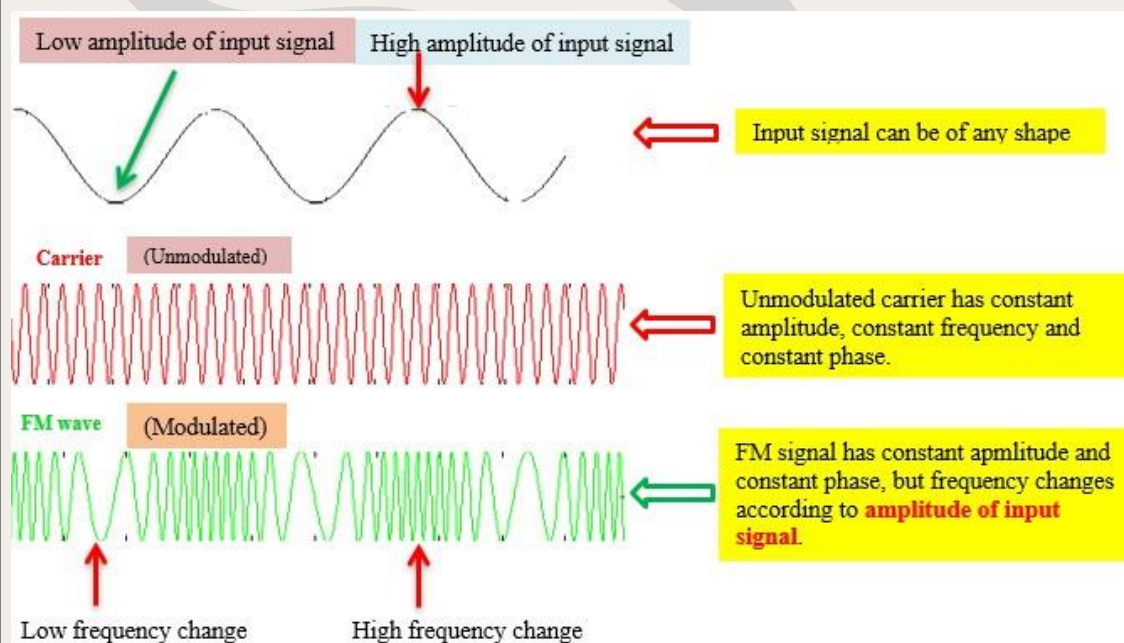


Figure 1.1 Frequency modulated wave

The modulating information signal is a low-frequency sine wave. As the sine wave goes positive, the frequency of the carrier increases proportionately. The highest frequency occurs at the peak amplitude of the modulating signal. As the modulating signal amplitude decreases, the carrier frequency decreases. When the modulating signal is at zero amplitude, the carrier is at its center frequency point.

Frequency modulation equation

$$v = A \sin [\omega c t + (\Delta f / f_m) \sin \omega_m t]$$

$$= A \sin [\omega c t + m_f \sin \omega_m t]$$

A = Amplitude of the FM signal. Δf = Frequency deviation

m_f = Modulation Index of FM

$$m_f = \Delta f / f_m$$

m_f is called the modulation index of frequency modulation.

Example

A transmitter operates on a frequency of 915 MHz. The maximum FM deviation is ± 12.5 kHz. What are the maximum and minimum frequencies that occur during modulation?

$$915 \text{ MHz} = 915,000 \text{ kHz}$$

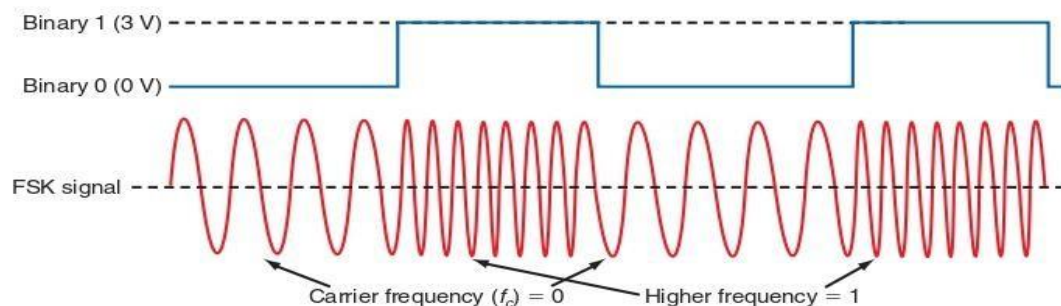
$$\text{Maximum deviation} = 915,000 + 12.5 = 915,012.5 \text{ kHz}$$

$$\text{Minimum deviation} = 915,000 - 12.5 = 914,987.5 \text{ kHz}$$

If the modulating signal is a pulse train or series of rectangular waves, e.g., serial binary data, then modulating signal has only two amplitudes, the carrier frequency, instead of having an infinite number of values, will only two values. This phenomenon is illustrated in Fig 1.2

For example, when the modulating signal is a binary 0, the carrier frequency is the center frequency value. When the modulating signal is a binary 1, the carrier frequency abruptly changes to a higher frequency level. The amount of the shift depends on the amplitude of the binary signal. This kind of modulation, called *frequency-shift keying (FSK)*, is widely used in the transmission of binary data in Bluetooth headsets, wireless speakers, and many forms of industrial wireless.

Figure 1-2 Frequency-modulating of a carrier with binary data produces FSK.



Principles of Phase Modulation

When the amount of phase shift of a constant-frequency carrier is varied in accordance with a modulating signal, the resulting output is a *phase modulation (PM)*

The output of the phase modulator is simply the carrier frequency that has been shifted in phase. This clearly illustrates that when a modulating signal is applied to a phase modulator, the output frequency changes only during the time that the amplitude of the modulating signal is varying. The maximum frequency deviation produced by a phase modulator occurs during the time when the modulating signal is changing at its most rapid rate

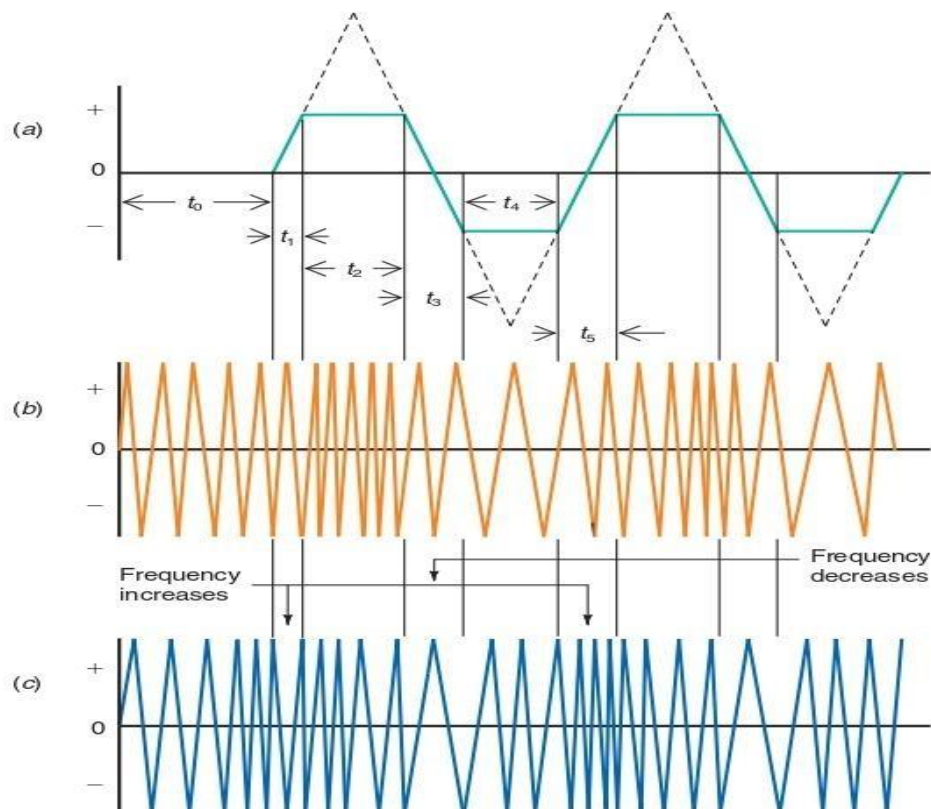


Figure 1i-3 A frequency shift occurs in PM only when the modulating signal amplitude varies. (a) Modulating signal. (b) FM signal. (c) PM signal.

Converting PM to FM:

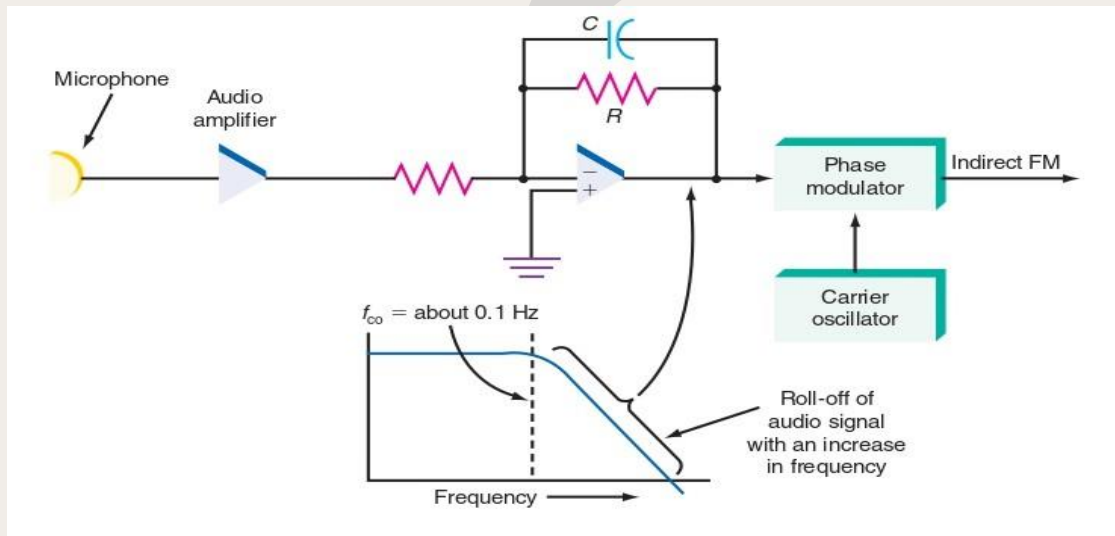


Figure 1.4 Converting PM to FM

To make PM compatible with FM, the deviation produced by frequency variations in the modulating signal must be compensated. This can be done by passing the message signal through a low-pass RC network, as shown Fig.. This low-pass filter, called a *frequency-correcting network*, *predistorter*, or *1/f filter*, causes the higher modulating frequencies to be attenuated. Although the higher modulating frequencies produce a greater rate of change and thus a greater frequency deviation, this is offset by the lower amplitude of the modulating signal, which produces less phase shift and thus less frequency deviation.

The predistorter compensates for the excess frequency deviation caused by higher modulating frequencies. The result is an output that is the same as an FM signal. The FM produced by a phase modulator is called *indirect FM*.

Phase-Shift Keying

PM is also used with binary signals, as shown in Fig. 1-5. When the binary modulating signal is 0 V, or binary 0, the PM signal is simply the carrier frequency. When a binary 1 voltage level occurs, the modulator, which is a phase shifter, simply changes the phase of the carrier, not its frequency. In Fig. 1-5 the phase shift is 180° . Each time the signal changes from 0 to 1 or 1 to 0, there is a 180° phase shift. The PM signal is still the carrier frequency, but the phase has been changed with respect to the original carrier with a binary 0 input. The process of phase-modulating a carrier with binary data is called *phase-shift keying (PSK)* or *binary phase-shift keying (BPSK)*. The PSK signal shown in Fig. 1-5 uses a 180° phase shift from a reference. The PSK signal has a constant frequency, but the phase of the signal from some reference changes as the binary modulating signal occurs.

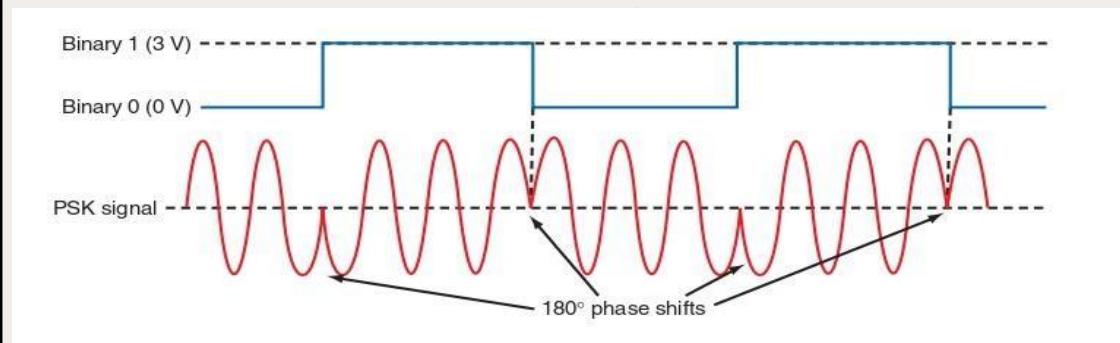


Figure 1.5 Phase shift keying

Modulation Index and Sidebands

When a constant-frequency sine wave modulates a carrier, two side frequencies are produced. The side frequencies are the sum and difference of the carrier and the modulating frequency.

In FM and PM, as in AM, sum and difference sideband frequencies are produced. In addition, a large number of pairs of upper and lower sidebands are generated.

As a result, the spectrum of an FM or a PM signal is usually wider than that of an equivalent AM signal.

It is also possible to generate a special narrowband FM signal whose bandwidth is only slightly wider than that of an AM signal.

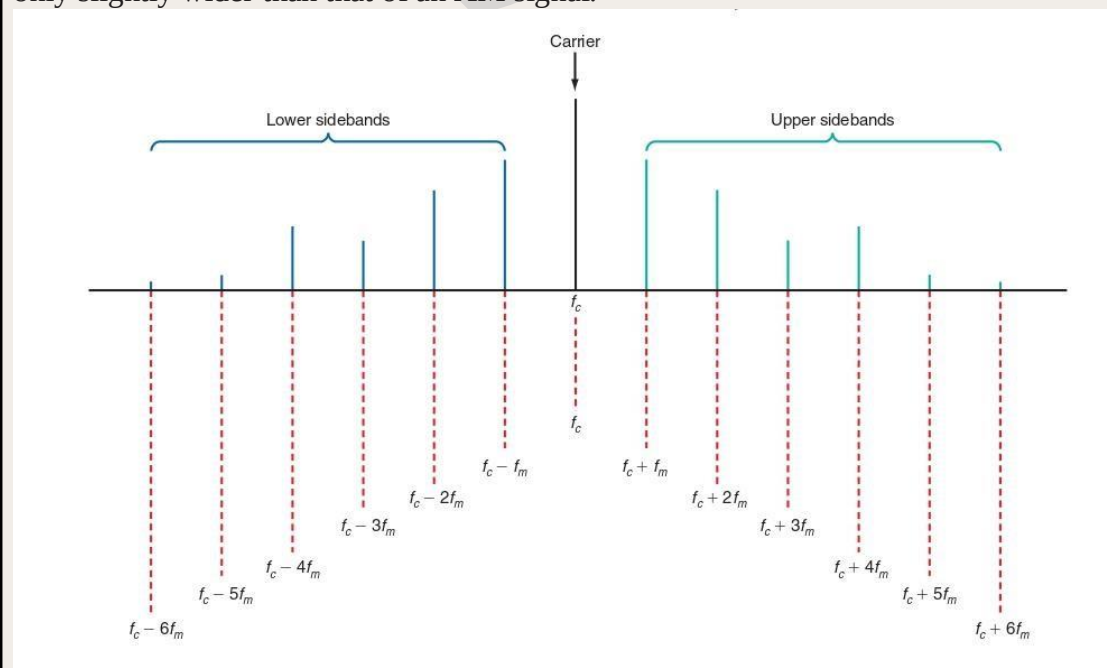


Figure 1.6 Frequency spectrum of FM

Fig. 1-6 shows the frequency spectrum of a typical FM signal produced by modulating a carrier with a single-frequency sine wave. Note that the sidebands are spaced from the carrier f_c and from one another by a frequency equal to the modulating frequency f_m .

If the modulating frequency is 1 kHz, the first pair of sidebands is above and below the carrier by 1000 Hz (1 kHz). The second pair of sidebands is above and below the carrier by 2 3 1000 Hz 5 2000 Hz, or 2 kHz, and so on.

The amplitudes of the sidebands vary. If each sideband is assumed to be a sine wave, with a frequency and an amplitude as indicated in Fig. and all the sine waves are added, then the FM signal producing them will be created.

As the amplitude of the modulating signal varies, the frequency deviation changes. The number of sidebands produced, and their amplitude and spacing, depends on the frequency deviation and modulating frequency. FM signal has constant amplitude. Since an FM signal is a summation of the sideband frequencies, the sideband amplitudes must vary with frequency deviation and modulating frequency if their sum is to produce a constant-amplitude but variable-frequency FM signal.

Modulation Index

The ratio of the frequency deviation to the modulating frequency is known as the **modulation index**

$$m_f = \frac{f_d}{f_m}$$

where f_d is the frequency deviation and f_m is the modulating frequency. Sometimes the lowercase Greek letter delta (δ) is used instead of f_d to represent deviation; then

$$m_f = \delta / f_m.$$

For example, if the maximum frequency deviation of the carrier is ± 12 kHz and the maximum modulating frequency is 2.5 kHz, the modulation index is $m_f = 12/2.5 = 4.8$.

In most communication systems using FM, maximum limits are put on both the frequency deviation and the modulating frequency. For example, in standard FM broadcasting, the maximum permitted frequency deviation is 75 kHz and the maximum permitted modulating frequency is 15 kHz. This produces a modulation index of $m_f = 75/15 = 5$.

When the maximum allowable frequency deviation and the maximum modulating frequency are used in computing the modulation index, m_f is known as the *deviation ratio*.

Example

What is the deviation ratio of TV sound if the maximum deviation is 25 kHz and the maximum modulating frequency is 15 kHz?

$$m_f = \frac{f_d}{f_m} = \frac{25}{15} = 1.667$$

Bessel Functions

The number and amplitudes of the significant sidebands can be determined by solving the basic equation of an FM signal.

FM signal is given by

$$v_{\text{FM}} = V_c \sin [2\pi f_c t + m_f \sin (2\pi f_m t)],$$

where v_{FM} is the instantaneous value of the FM signal and m_f is the modulation index. The term whose coefficient is m_f is the phase angle of the carrier. Note that this equation expresses the phase angle in terms of the sine wave modulating signal. This equation is solved with a complex mathematical process known as *Bessel functions*, but the result is as follows:

$$\begin{aligned} v_{\text{FM}} = V_c \{ & J_0(\sin \omega_c t) + J_1[\sin(\omega_c + \omega_m)t - \sin(\omega_c - \omega_m)t] \\ & + J_2[\sin(\omega_c + 2\omega_m)t + \sin(\omega_c - 2\omega_m)t] \\ & + J_3[\sin(\omega_c + 3\omega_m)t - \sin(\omega_c - 3\omega_m)t] \\ & + J_4[\sin(\omega_c + 4\omega_m)t + \sin(\omega_c - 4\omega_m)t] \\ & + J_5[\sin \dots] + \dots \} \end{aligned}$$

where $\omega_c = 2\pi f_c$ = carrier frequency

$\omega_m = 2\pi f_m$ = modulating signal frequency

V_c = peak value of unmodulated carrier

The FM wave is expressed as a composite of sine waves of different frequencies and amplitudes that, when added, give an FM time-domain signal.

The first term is the carrier with an amplitude given by a J_n coefficient, in this case J_0 . The next term represents a pair of upper and lower side frequencies equal to the sum and difference of the carrier and modulating signal frequency. The amplitude of these side frequencies is J_1 . The next term is another pair of side frequencies. The other terms represent additional side frequencies spaced from one another by an amount equal to the modulating signal frequency

The amplitudes of the sidebands are determined by the J_n coefficients, which are, in turn, determined by the value of the modulation index. These amplitude coefficients are computed by using the expression

$$J_n(m_f) = \left(\frac{m_f}{2^n n!} \right)^n \left[1 - \frac{(m_f)^2}{2(2n+2)} + \frac{(m_f)^4}{2 \cdot 4(2n+2)(2n+4)} - \frac{(m_f)^6}{2 \cdot 4 \cdot 6(2n+2)(2n+4)(2n+6)} + \dots \right]$$

where ! = factorial

n = sideband number (1, 2, 3, etc.)

$n = 0$ is the carrier

$$m_f = \frac{f_d}{f_m} = \text{frequency deviation}$$

Example

What is the maximum modulating frequency that can be used to achieve a modulation index of 2.2 with a deviation of 7.48 kHz?

$$f_m = \frac{f_d}{m_f} = \frac{7480}{2.2} = 3400 \text{ Hz} = 3.4 \text{ kHz}$$

FM signal spectrum for different modulation index

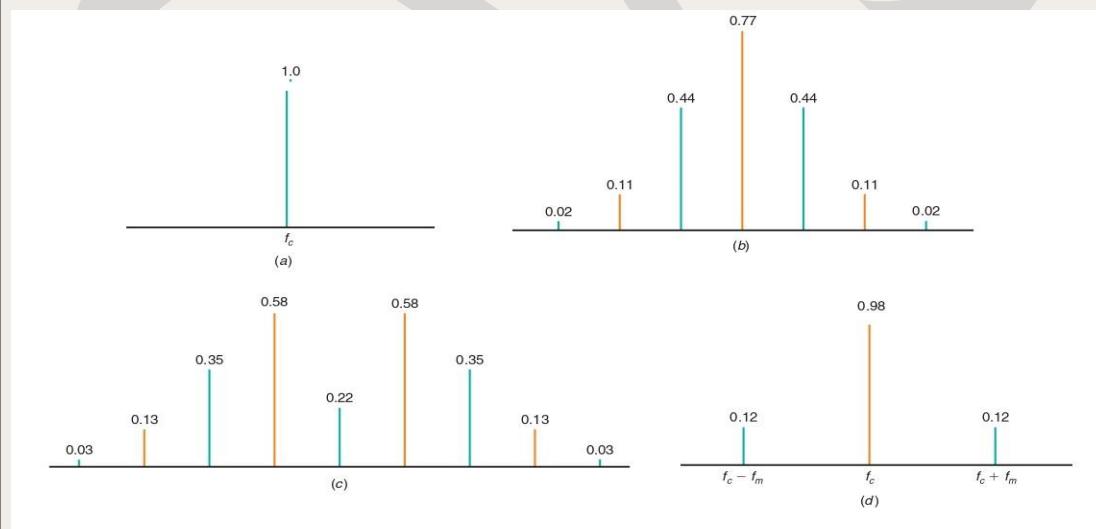


Figure 1.7 Examples of FM signal spectra. (a) Modulation index of 0 (no modulation or sidebands). (b) Modulation index of 1. (c) Modulation index of 2. (d) Modulation index of 0.25 (NBFM).

Fig. 1-7 shows several examples of an FM signal spectrum with different modulation indexes.

The unmodulated carrier in Fig. 1-7(a) has a relative amplitude of 1.0. With no modulation, all the power is in the carrier. With modulation, the carrier amplitude decreases while the amplitudes of the various sidebands increase.

FM Signal Bandwidth

The higher the modulation index in FM, the greater the number of significant sidebands and the wider the bandwidth of the signal. When spectrum conservation is necessary, the bandwidth of an FM signal can be deliberately restricted by putting an upper limit on the modulation index.

The total bandwidth of an FM signal can be determined by knowing the modulation index.

For example, assume that the highest modulating frequency of a signal is 3 kHz and the maximum deviation is 6 kHz. This gives a modulation index of $m_f = 6 \text{ kHz}/3 \text{ kHz} = 2$.

The bandwidth can then be determined with the simple formula

$$BW = 2f_m N$$

where N is the number of significant sidebands in the signal. According to this formula, the bandwidth of our FM signal is

$$BW = 2(3 \text{ kHz})(4) = 24 \text{ kHz}$$

Another way to determine the bandwidth of an FM signal is to use *Carson's rule*.

$$BW = 2[f_{d(\max)} + f_{m(\max)}]$$

Example

What is the maximum bandwidth of an FM signal with a deviation of 30 kHz and a maximum modulating signal of 5 kHz as determined by (a) Fig. 5-8 and (b) Carson's rule?

a. $m_f = \frac{f_d}{f_m} = \frac{30 \text{ kHz}}{5 \text{ kHz}} = 6$

Fig. 5-8 shows nine significant sidebands spaced 5 kHz apart for $m_f = 6$.

$$BW = 2f_m N = 2(5 \text{ kHz}) 9 = 90 \text{ kHz}$$

b. $BW = 2[f_{d(\max)} + f_{m(\max)}]$
 $= 2(30 \text{ kHz} + 5 \text{ kHz})$
 $= 2(35 \text{ kHz})$

$$BW = 70 \text{ kHz}$$

Noise Suppression Effects of FM

Noise is interference generated by lightning, motors, automotive ignition systems, and any power line switching that produces transient signals. Such noise is typically sharp spikes of voltage with very high frequencies. They add to a signal and interfere with it. The potential effect of such noise on an FM signal is shown in Fig. 1-8. If the noise signals were strong enough, they could completely obliterate the information signal.

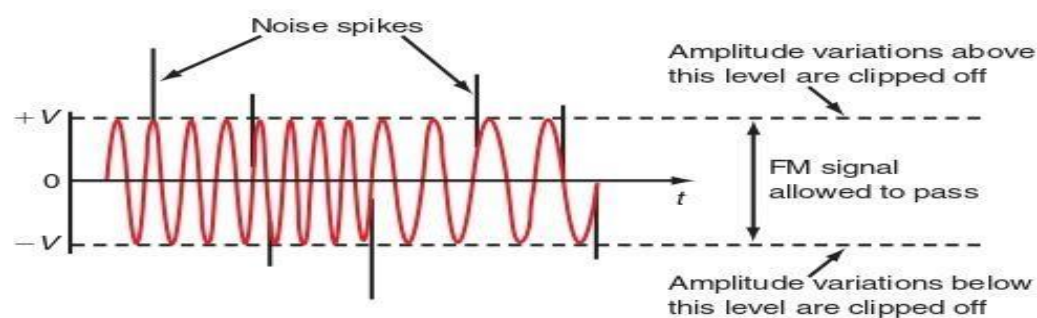


Figure 1.8 Noise spikes

FM signals, however, have a constant modulated carrier amplitude, and FM receivers

contain limiter circuits that deliberately restrict the amplitude of the received signal. Any amplitude variations occurring on the FM signal are effectively clipped off, as shown in Fig. 1.8. This does not affect the information content of the FM signal, since it is contained solely within the frequency variations of the carrier. Because of the clipping action of the limiter circuits, noise is almost completely eliminated. Even if the peaks of the FM signal itself are clipped or flattened and the resulting signal is distorted, no information is lost.

One of the primary benefits of FM over AM is its superior noise immunity. The process of demodulating or recovering an FM signal actually suppresses noise and improves the signal-to-noise ratio.

Noise and Phase Shift

The noise amplitude added to an FM signal introduces a small frequency variation, or phase shift, which changes or distorts the signal. Fig. 1-9 shows how this works. The carrier signal is represented by a fixed-length (amplitude) phasor S . The noise is usually a short duration pulse containing many frequencies at many amplitudes and phases according to Fourier theory. To simplify the analysis, however, we assume a single high-frequency noise signal varying in phase.

In Fig. 1-9(a), this noise signal is represented as a rotating phasor N . The composite signal of the carrier and the noise, labeled C , is a phasor whose amplitude is the phasor sum of the signal and noise and a phase angle shifted from the carrier by an amount ϕ .

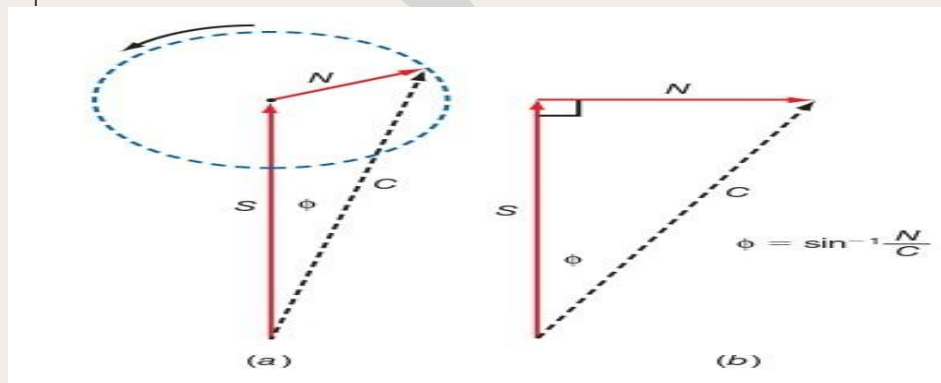


Figure 1.9 How noise introduces phase shift

$$\phi = \sin^{-1} \frac{N}{C}$$

It is possible to determine just how much of a frequency shift a particular phase shift produces by using the formula

$$\delta = \phi(f_m)$$

where δ = frequency deviation produced by noise
 ϕ = phase shift, rad
 f_m = frequency of modulating signal

Example

The input to an FM receiver has an S/N of 2.8. The modulating frequency is 1.5 kHz. The maximum permitted deviation is 4 kHz. What are (a) the frequency deviation

$$\text{a. } \phi = \sin^{-1} \frac{N}{S} = \sin^{-1} \frac{1}{2.8} = \sin^{-1} 0.3571 = 20.92^\circ \quad \text{or} \quad 0.3652 \text{ rad}$$

$$\delta = \phi(f_m) = (0.3652)(1.5 \text{ kHz}) = 547.8 \text{ Hz}$$

Preemphasis

Noise *can* interfere with an FM signal, and particularly with the high-frequency components of the modulating signal. Since noise is primarily sharp spikes of energy, it contains a lot of harmonics and other high-frequency components. These frequencies can be larger in amplitude than the high-frequency content of the modulating signal, causing frequency distortion that can make the signal unintelligible.

Most of the content of a modulating signal, particularly voice, is at low frequencies. In voice communication systems, the bandwidth of the signal is limited to about 3 kHz, which permits acceptable intelligibility.

In contrast, musical instruments typically generate signals at low frequencies but contain many high-frequency harmonics that give them their unique sound and must be passed if that sound is to be preserved. Thus a wide bandwidth is needed in high-fidelity systems. Since the high-frequency components are usually at a very low level, noise can obliterate them.

To overcome this problem, most FM systems use a technique known as *preemphasis* that helps offset high-frequency noise interference. At the transmitter, the modulating signal is passed through a simple network that amplifies the high-frequency components more than the low-frequency components.

The simplest form of such a circuit is a simple high-pass filter of the type shown in Fig. 1-10(a).

Frequency Modulation (FM), **pre-emphasis** is a technique used to improve the signal-to-noise ratio (SNR) at higher frequencies. This process boosts the amplitude of the higher-frequency components of the signal before transmission. It is commonly used in FM broadcasting to ensure that the high-frequency signals are transmitted with sufficient strength, which helps to reduce the impact of noise or interference on the received signal.

The preemphasis circuit also has an upper break frequency f_u , at which the signal enhancement flattens out [see Fig. 1-10(b)], which is computed with the formula

$$f_u = \frac{R_1 + R_2}{2\pi R_1 R_2 C}$$

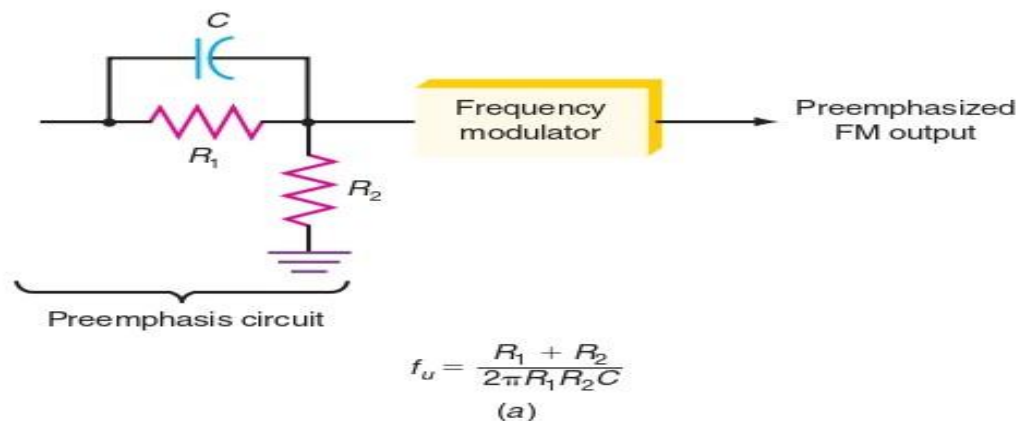


Figure 1.10(a) Preemphasis circuit.

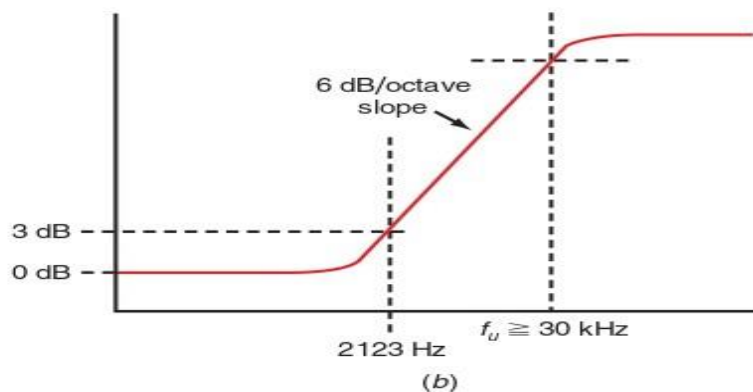


Figure 1.10(b) Preemphasis curve

Preemphasis is used to

- **Improves Signal-to-Noise Ratio (SNR):** Higher frequencies are more prone to attenuation and noise, so pre-emphasizing them helps to maintain the overall quality of the signal at the receiver.
- **Compensates for Transmission Losses:** Transmission systems (especially in analog FM) tend to lose higher frequencies more than lower ones. Pre-emphasis compensates for this loss.
- **Enhances Audio Clarity:** In broadcasting, higher frequencies often contain important details like sibilance in speech or sharp musical tones. Emphasizing these frequencies improves overall clarity.

Deemphasis

De-emphasis: On the receiver side a complementary process called **de-emphasis** is applied. It attenuates the higher frequencies by the same amount as they were boosted during pre-emphasis, effectively restoring the original balance of the signal and reducing noise to return the frequency response to its normal, “flat” level, a *deemphasis circuit*, a simple low-pass filter with a time constant of 75 μs , is used at the receiver which is shown in Fig. 1-10(c)].

Signals above its cutoff frequency are attenuated at the rate of 6 dB per octave. The response curve is shown in Fig. 1-10(d). As a result, the preemphasis at the transmitter is exactly offset by the deemphasis circuit in the receiver, providing a flat frequency response. The combined effect of preemphasis and deemphasis is to increase the signal-to-noise ratio for the high-frequency components during transmission so that they will be stronger and not masked by noise. Fig. 1-10(e) shows the overall effect of preemphasis and deemphasis.

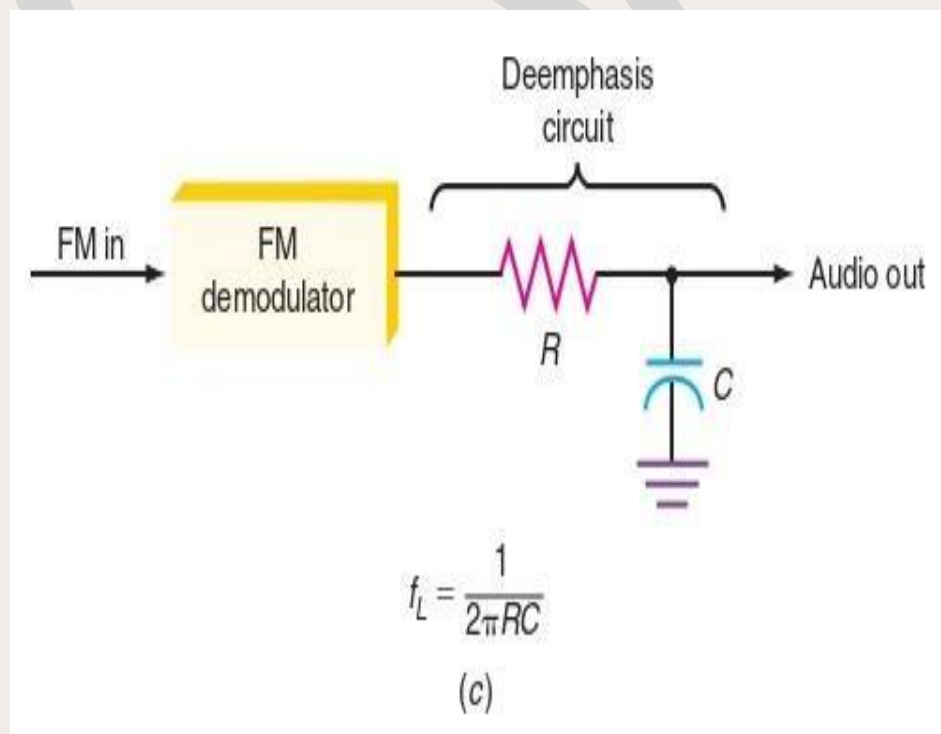


Figure 1.10(c) Deemphasis circuit

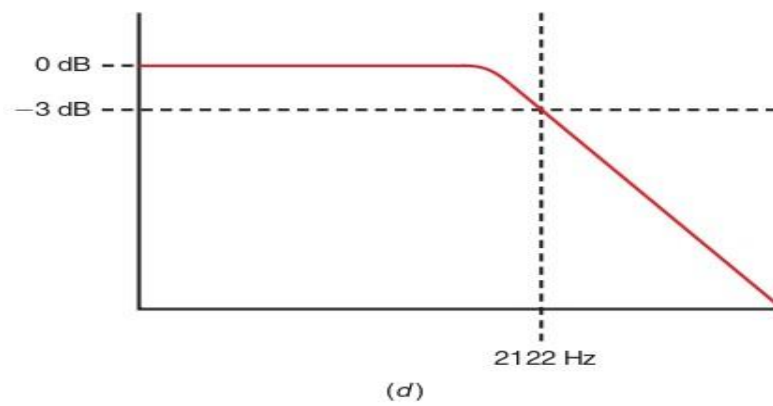


Figure 1.10(d) Deemphasis curve

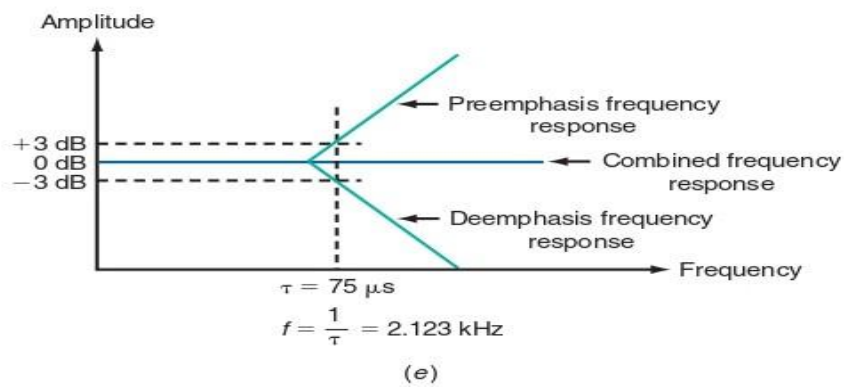


Figure 1.10(e) Combined frequency response

Frequency Modulation Versus Amplitude Modulation:

Advantages of FM:

Noise Immunity. The main benefit of FM over AM is its superior immunity to noise, made possible by the clipper limiter circuits in the receiver, which effectively strip off all the noise variations, leaving a constant-amplitude FM signal. Although clipping does not result in total recovery in all cases, FM can nevertheless tolerate a much higher noise level than AM for a given carrier amplitude.

Capture Effect. Another major benefit of FM is that interfering signals on the same frequency are effectively rejected. Because of the amplitude limiters and the demodulating methods used by FM receivers, a phenomenon known as the *capture effect* takes place when two or more FM signals occur simultaneously on the same frequency. If one signal is more than twice the amplitude of the other, the stronger signal captures the channel, totally eliminating the weaker signal. With modern receiver circuitry, a difference in signal amplitudes of only 1 dB is usually sufficient to produce the capture effect. In contrast, when two AM signals occupy the same frequency, both signals are generally heard, regardless of their relative signal strengths.

Although the capture effect prevents the weaker of two FM signals from being heard, when two stations are broadcasting signals of approximately the same amplitude, first one may be captured and then the other. This can happen, e.g., when a driver moving along a highway is listening to a clear broadcast on a particular frequency. At some point, the driver may suddenly hear the other broadcast, completely losing the first, and then, just as suddenly, hear the original broadcast again. Which one dominates depends on where the car is and on the relative signal strengths of the two signals.

Transmitter Efficiency.

FM signals have a constant amplitude, and it is therefore not necessary to use linear amplifiers to increase their power level. In fact, FM signals are always generated at a lower level and then amplified by a series of class C amplifiers to increase their power. The result is greater use of available power because of the high level of efficiency of class C amplifiers. Even more efficient class D, E, or F amplifiers are also used in FM or PM equipment.

Disadvantages of FM

Excessive Spectrum Use. Perhaps the greatest disadvantage of FM is that it simply uses too much spectrum space. The bandwidth of an FM signal is, wider than that of an AM signal transmitting similar information. Although it is possible to keep the modulation index low to minimize bandwidth, reducing the modulation index also reduces the noise immunity of an FM signal. Most FM communication work is done at the VHF, UHF, and microwave frequencies.

Circuit Complexity. One major disadvantage of FM in the past involved the complexity of the circuits used for frequency modulation and demodulation in comparison with the simple circuits used for amplitude modulation and demodulation. Today, this disadvantage has almost disappeared because of the use of integrated circuits.

Although the ICs used in FM transmission are still complex, they require very little effort to use and their price is just as low as those of comparable AM circuits. Since the trend in electronic communication is toward higher and higher frequencies and because ICs are so cheap and easy to use, FM and PM have become by far the most widely used modulation method in electronic communication today.

Frequency Modulators

A *frequency modulator* is a circuit that varies *carrier frequency* in accordance with the modulating signal. The carrier is generated by either an *LC* or a crystal oscillator circuit.

Varactor is used in frequency modulators whose capacitance will change in response to the modulating signal. Also known as a voltage variable capacitor, variable capacitance diode, or varicap, this device is basically a semiconductor junction diode operated in a reverse-bias mode.

Varactor Operation

A junction diode is created when P- and N-type semiconductors are formed. Some electrons in the N-type material drift over into the P-type material and neutralize the holes there [see Fig. 1-11(a)], forming a thin area called the *depletion region*, where there are no free carriers, holes, or electrons.

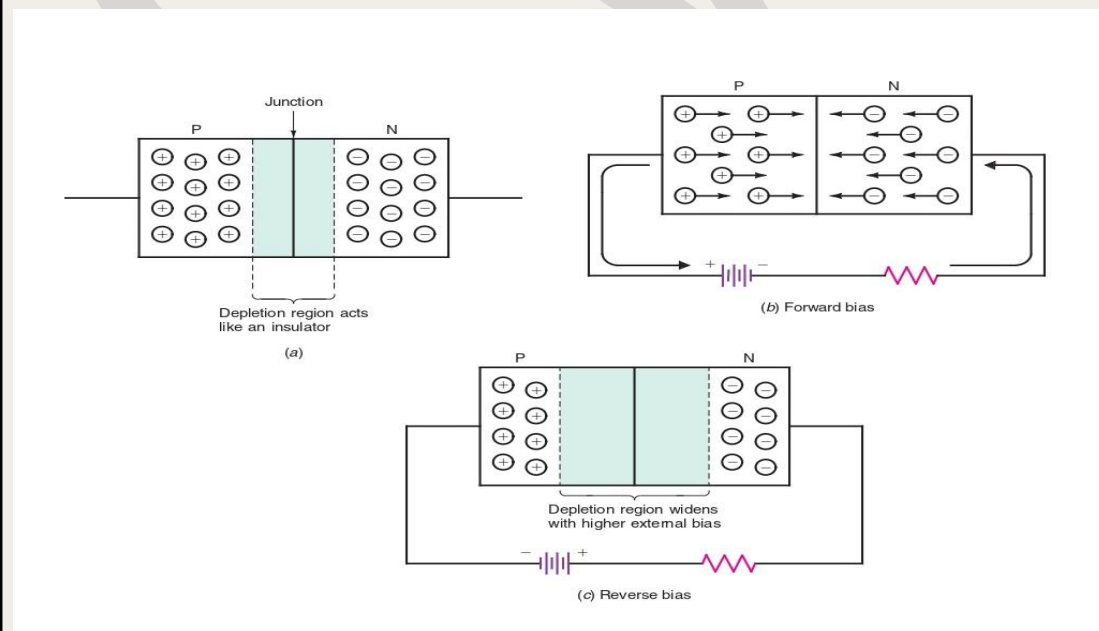


Figure 1.11 Depletion region in diode

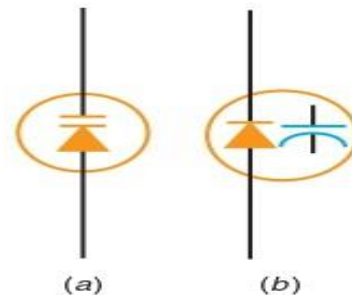
The depletion region acts as a thin insulator that prevents current from flowing through the device. If a forward bias is applied to the diode, it will conduct. The external potential forces the holes and electrons toward the junction, where they combine and cause a continuous current inside the diode as well as externally. The depletion layer simply disappears [Fig. 1-11(b)].

If an external reverse bias is applied to the diode, as in Fig. 1-11(c), no current will flow. The bias increases the width of the depletion layer, with the amount of increase depending on the amount of the reverse bias. The higher the reverse bias, the wider the depletion layer and the less chance for current flow

A reverse-biased junction diode acts as a small capacitor. The P- and N-type materials act as the two plates of the capacitor, and the depletion region acts as the dielectric. With all the active current carriers (electrons and holes) neutralized in the depletion region, it functions just as an insulating material. The width of the depletion layer determines the width of the dielectric and, therefore, the amount of capacitance. If the reverse bias is high, the depletion region will be wide and the dielectric will cause the plates of the capacitor to be widely spaced, producing a low capacitance. Decreasing the amount of reverse bias narrows the depletion region; the plates of the capacitor are effectively closer together, producing a higher capacitance.

All junction diodes exhibit variable capacitance as the reverse bias is changed. Varactors are designed to optimize this particular characteristic, so that the capacitance variations are as wide and linear as possible. The symbols used to represent varactor diodes are shown in Fig. 1-12.

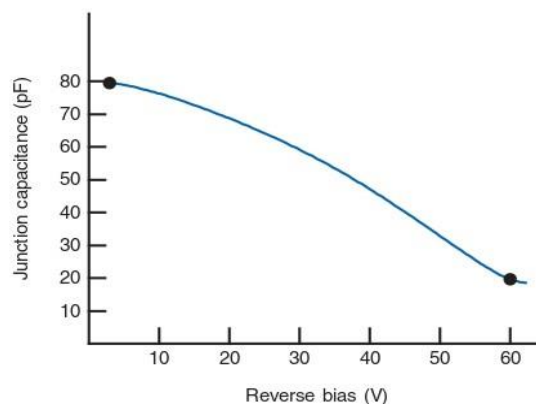
Figure 1.12 Schematic symbols of a varactor diode.



Varactors are made with a wide range of capacitance values, most units having a nominal capacitance in the 1- to 200-pF range. The capacitance variation range can be as high as 12 :1.

Fig. 1-13 shows the curve for a typical diode. A maximum capacitance of 80 pF is obtained at 1 V. With 60 V applied, the capacitance drops to 20 pF, a 4 :1 range.

Figure 1.13 Capacitance versus reverse junction voltage for a typical varactor.



Varactor Modulators

Figure 1.14 A direct-frequency-modulated carrier oscillator using a varactor diode.

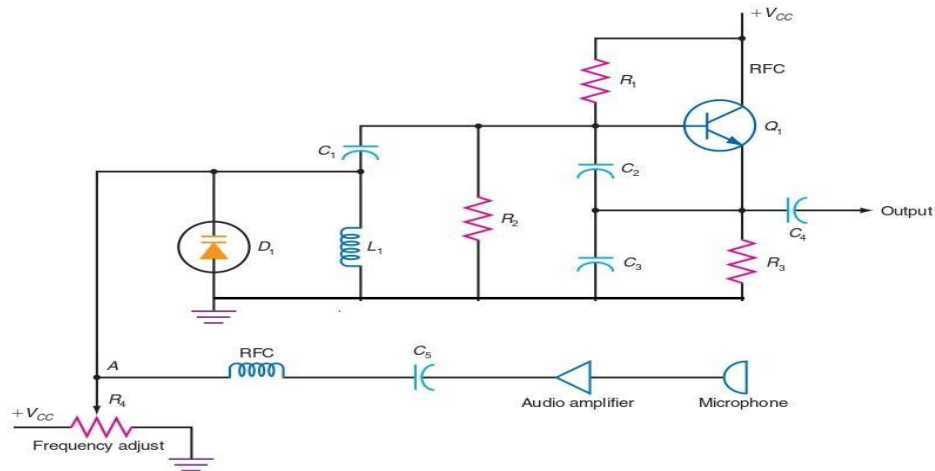


Fig. 1-14, a carrier oscillator for a transmitter, shows the basic concept of a varactor frequency modulator. The capacitance of varactor diode $D1$ and $L1$ forms the parallel-tuned circuit of the oscillator. The value of $C1$ is made very large at the operating frequency so that its reactance is very low. As a result, $C1$ connects the tuned circuit to the oscillator circuit. Also $C1$ blocks the dc bias on the base of $Q1$ from being shorted to ground through $L1$. The values of $L1$ and $D1$ fix the center carrier frequency.

The capacitance of $D1$ is controlled in two ways, through a fixed dc bias and by the modulating signal.

In the figure the bias on $D1$ is set by the voltage divider potentiometer $R4$. Varying $R4$ allows the center carrier frequency to be adjusted over a narrow range. The modulating signal is applied through $C5$ and the radio frequency choke (RFC); $C5$ is a blocking capacitor that keeps the dc varactor bias out of the modulating-signal circuits. The reactance of the RFC is high at the carrier frequency to prevent the carrier signal from getting back into the audio modulating-signal circuits.

The modulating signal derived from the microphone is amplified and applied to the modulator. As the modulating signal varies, it adds to and subtracts from the fixed-bias voltage. Thus, the effective voltage applied to $D1$ causes its capacitance to vary. This, in turn, produces the desired deviation of the carrier frequency. A positive-going signal at point A adds to the reverse bias, decreasing the capacitance and increasing the carrier frequency. A negative-going signal at A subtracts from the bias, increasing the capacitance and decreasing the carrier frequency.

Example

The value of capacitance of a varactor at the center of its linear range is 40 pF. This varactor will be in parallel with a fixed 20-pF capacitor. What value of inductance should be used to resonate this combination to 5.5 MHz in an oscillator? Total capacitance $C_T = 40 + 20 = 60$ pF.

$$\begin{aligned} f_0 &= 5.5 \text{ MHz} = \frac{1}{2\pi\sqrt{LC_T}} \\ L &= \frac{1}{(2\pi f)^2 C_T} = \frac{1}{(6.28 \times 5.5 \times 10^6)^2 \times 60 \times 10^{-12}} \\ L &= 13.97 \times 10^{-6} \text{ H} \quad \text{or} \quad 14 \mu\text{H} \end{aligned}$$

Frequency-Modulating a Crystal Oscillator

It is possible to vary the frequency of a crystal oscillator by changing the value of capacitance in series or in parallel with the crystal. Fig. 1-15 shows a typical crystal oscillator. When a small value of capacitance is connected in series with the crystal, the crystal frequency can be “pulled” slightly from its natural resonant frequency. By making the series capacitance a varactor diode, frequency modulation of the crystal oscillator can be achieved. The modulating signal is applied to the varactor diode D_1 , which changes the oscillator frequency. It is important to note that only a very small frequency deviation is possible with frequency-modulated crystal oscillators. Rarely can the frequency of a crystal oscillator be changed more than several hundred hertz from the nominal crystal value. The resulting deviation may be less than the total deviation desired. For example, to achieve a total frequency shift of 75 kHz, which is necessary in commercial FM broadcasting, other techniques must be used. In NBFM communication systems, the narrower deviations are acceptable. Although it is possible to achieve a deviation of only several hundred cycles from the crystal oscillator frequency, the total deviation can be increased by using frequency multiplier circuits after the carrier oscillator.

A *frequency multiplier circuit* is one whose output frequency is some integer multiple of the input frequency. A frequency multiplier that multiplies a frequency by 2 is called a *doubler*, a frequency multiplier circuit that multiplies an input frequency by 3 is called a *tripler*, and so on.

Frequency multipliers can also be cascaded. When the FM signal is applied to a frequency multiplier, both the carrier frequency of operation and the amount of deviation are increased.

Fig. 1-16 shows how frequency multipliers increase carrier frequency and deviation.

Figure 1.15 Frequency modulation of a crystal oscillator with a VVC.

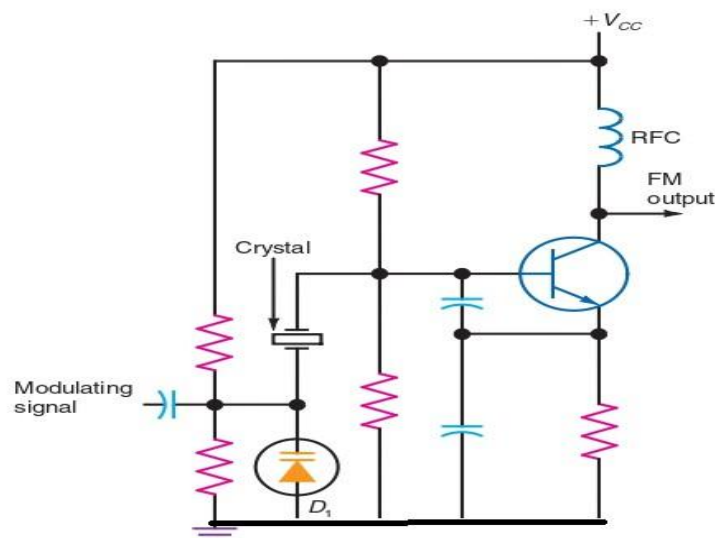
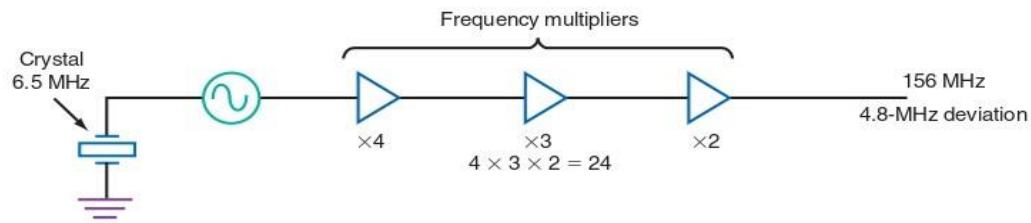


Figure 1.16 How frequency multipliers increase carrier frequency and deviation.



Voltage-Controlled Oscillators

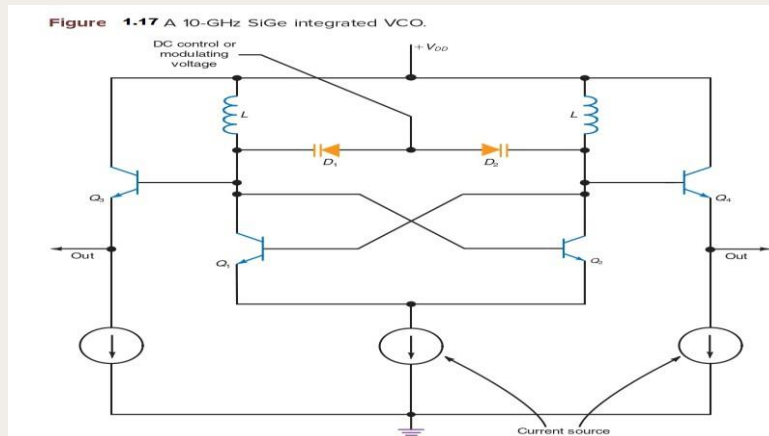
Oscillators whose frequencies are controlled by an external input voltage are generally referred to as *voltage-controlled oscillators (VCOs)*. *Voltage-controlled crystal oscillators* are generally referred to as *VCOs*.

Although VCOs for VHF, UHF, and microwaves are still implemented with discrete components, more and more they are being integrated on a single chip of silicon along with other transmitter or receiver circuits.

An example of such a VCO is shown in Fig. 1-17. This circuit uses silicon-germanium (SiGe) bipolar transistor to achieve an operating frequency centered near 10 GHz. The oscillator uses cross-coupled transistors Q1 and Q2. The signal is a sine wave whose frequency is set by the collector inductances and varactor capacitances.

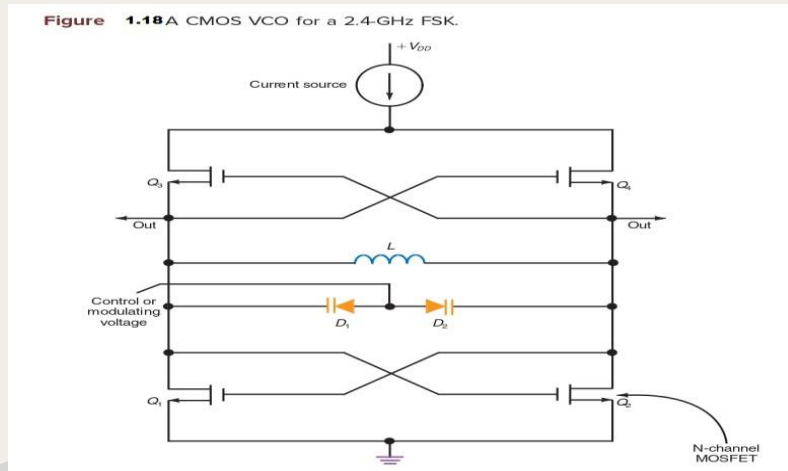
The modulating voltage, usually a binary signal to produce FSK, is applied to the junction of D1 and D2. Two complementary outputs are available from the emitter followers Q3 and Q4. In this circuit, the inductors are actually tiny spirals of aluminum (or copper) inside the chip.

The varactors are reverse-biased diodes that function as variable capacitors. The tuning range is from 9.953 to 10.66 GHz.



A CMOS version of the VCO is shown in Fig. 1-18. This circuit also uses a crosscoupled LC resonant circuit design and operates in the 2.4- to 2.5-GHz range. Variations of it are used in Bluetooth transceivers and wireless LAN applications. There are also many different types of lower-frequency VCOs in common use, including IC VCOs using RC multivibrator-type oscillators whose frequency can be controlled over a wide range by an ac or dc input voltage.

Figure 1.18A CMOS VCO for a 2.4-GHz FSK.



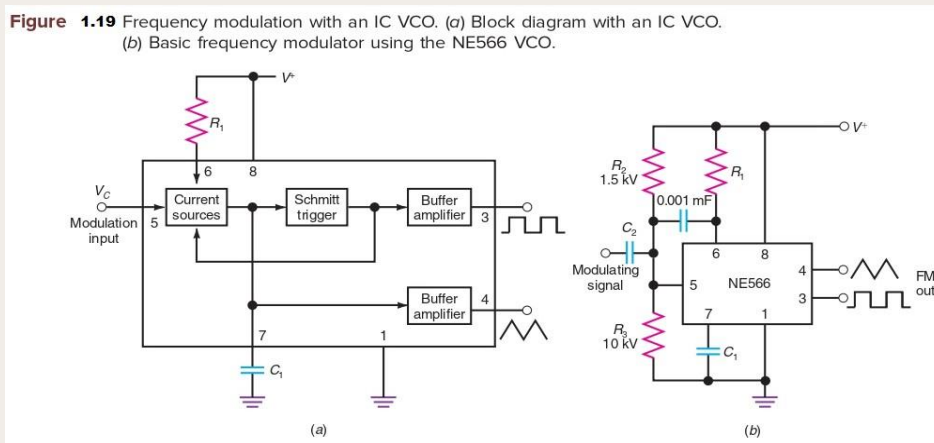
Frequency modulation IC VCO

Fig. 1-19(a) is a block diagram of one widely used IC VCO, the popular NE566. External resistor R_1 at pin 6 sets the value of current produced by the internal current sources. The current sources linearly charge and discharge external capacitor C_1 at pin 7. An external voltage V_C applied at pin 5 is used to vary the amount of current produced by the current sources. The Schmitt trigger circuit is a level detector that controls the current source by switching between charging and discharging when the capacitor charges or discharges to a specific voltage level.

A linear sawtooth of voltage is developed across the capacitor by the current source. This is buffered by an amplifier and made available at pin 4. The Schmitt trigger output is a square wave at the same frequency available at pin 3. If a sine wave output is desired, the triangular wave is usually filtered with a tuned circuit resonant to the desired carrier frequency.

A complete frequency modulator circuit using the NE566 is shown in Fig. 1-19(b). The current sources are biased with a voltage divider made up of R_2 and R_3 . The modulating signal is applied through C_2 to the voltage divider at pin 5. The 0.001- μF capacitor between pins 5 and 6 is used to prevent unwanted oscillations. The center carrier frequency of the circuit is set by the values of R_1 and C_1 . Carrier frequencies up to 1 MHz may be used with this IC. If higher frequencies and deviations are necessary, the outputs can be filtered or used to drive other circuits, such as a frequency multiplier. The modulating signal can vary the carrier frequency over nearly a 10 :1 range, making very large deviations possible. The deviation is linear with respect to the input amplitude over the entire range.

Figure 1.19 Frequency modulation with an IC VCO. (a) Block diagram with an IC VCO. (b) Basic frequency modulator using the NE566 VCO.



Frequency Demodulators

A demodulator is a device which extracts information carrying signal from the frequency modulated signal..

Circuits used to recover the original modulating signal from an FM transmission are called demodulators, detectors, or discriminators.

Slope Detectors

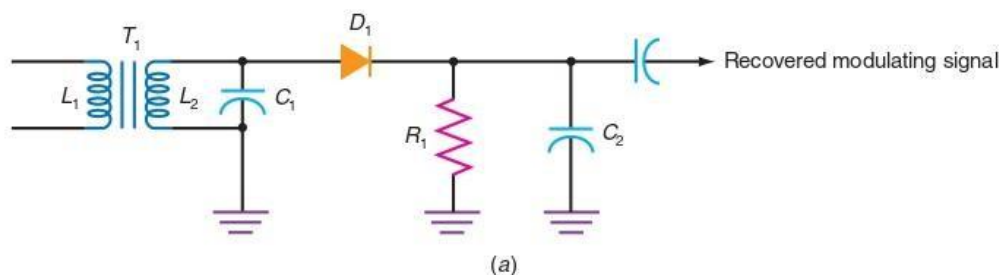
The slope detector, makes use of a tuned circuit and a diode detector to convert frequency variations to voltage variations. The basic circuit is shown in Fig. 1-20(a).

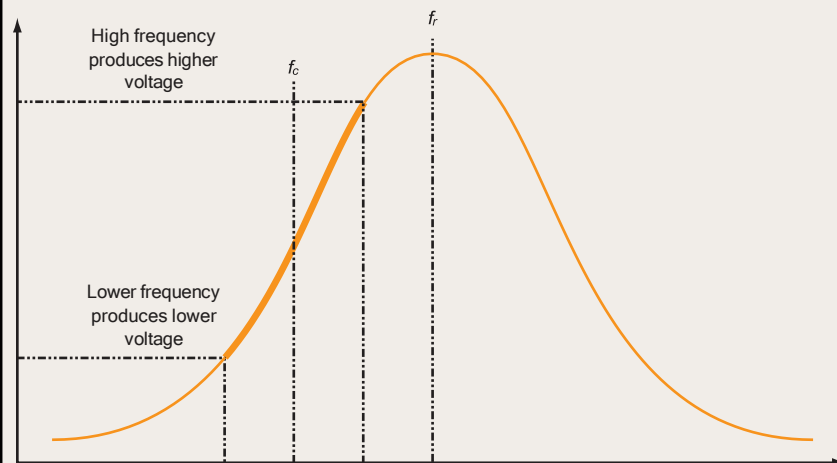
The FM signal is applied to transformer T_1 made up of L_1 and L_2 . Together L_2 and C_1 form a series resonant circuit. The signal voltage induced into L_2 appears in series with L_2 and C_1 and the output voltage is taken from across C_1 . The response curve of this tuned circuit is shown in Fig. 1-20(b).

At the resonant frequency f_r the voltage across C_1 reaches the peak value. At lower or higher frequencies, the voltage falls off. To use the circuit to detect or recover FM, the circuit is tuned so that the center or carrier frequency of the FM signals is approximately centered on the leading edge of the response curve, as shown in Fig. 6-12(b).

As the carrier frequency varies above and below its center frequency, the tuned circuit responds as shown in the figure. If the frequency goes lower than the carrier frequency, the output voltage across C_1 decreases. If the frequency goes higher, the output across C_1 goes higher. Thus, the ac voltage across C_1 is proportional to the frequency of the FM signal. The voltage across C_1 is rectified into dc pulses that appear across the load R_1 . These are filtered into a varying dc signal that is an exact reproduction of the original modulating signal. The main difficulty with slope detectors lies in tuning them so that the FM signal is correctly centered on the leading edge of the tuned circuit. In addition, the tuned circuit does not have a perfectly linear response. It is approximately linear over a narrow range, as Fig. 6-12(b) shows, but for wide deviations, amplitude distortion occurs because of the nonlinearity

Figure 1.20 Slope detector operation.

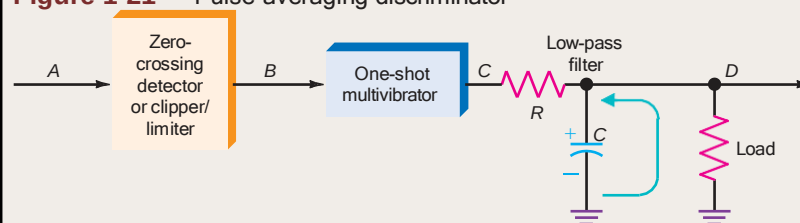




Pulse-Averaging Discriminators

A simplified block diagram of a *pulse-averaging discriminator* is illustrated in Fig. 1-21. The FM signal is applied to a zero-crossing detector or a clipper-limiter that generates a binary voltage-level change each time the FM signal varies from minus to plus or from plus to minus. The result is a rectangular wave containing all the frequency variations of the original signal but without amplitude variations. The FM square wave is then applied to a one-shot (monostable) multivibrator that generates a fixed-amplitude, fixed-width dc pulse on the leading edge of each FM cycle. The duration of the one shot is set so it is less than one-half the period of the highest frequency expected during maximum deviation. The one-shot output pulses are then fed to a simple RC low-pass filter that averages the dc pulses to recover the original modulating signal.

Figure 1-21 Pulse-averaging discriminator



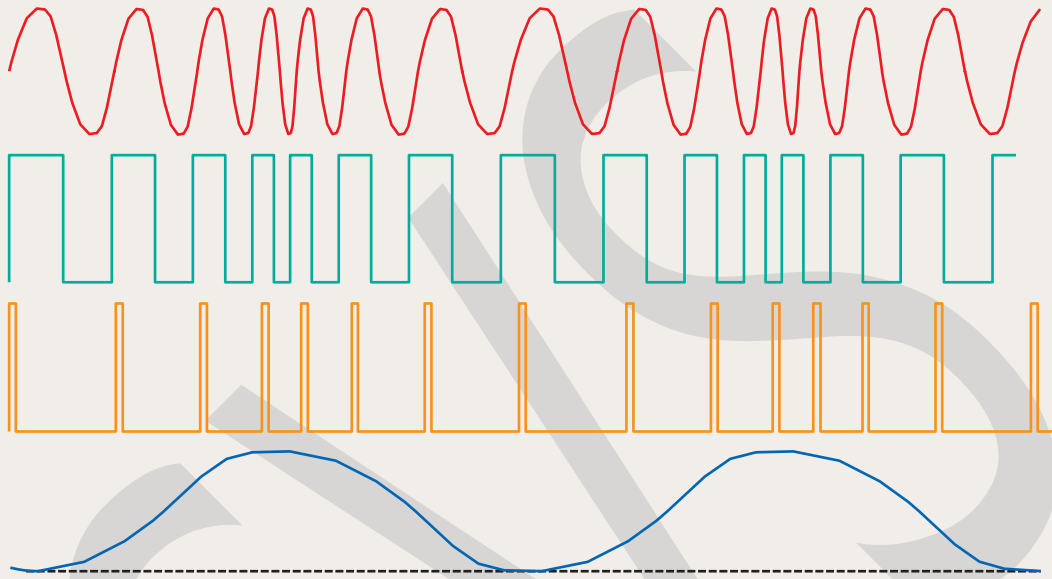


Figure 1-22 (a) FM input. (b) Output of zero-crossing detector. (c) Output of one shot (d) Output of discriminator (original modulating signal)

The waveforms for the pulse-averaging discriminator are illustrated in Fig. 1-22. At low frequencies, the one-shot pulses are widely spaced; at higher frequencies, they occur very close together. When these pulses are applied to the averaging filter, a dc output voltage is developed, the amplitude of which is directly proportional to the frequency deviation.

When a one-shot pulse occurs, the capacitor in the filter charges to the amplitude of the pulse. When the pulse turns off, the capacitor discharges into the load. If the RC time constant is high, the charge on the capacitor does not decrease much.

When the time interval between pulses is long, however, the capacitor loses some of its charge into the load, so the average dc output is low. When the pulses occur rapidly, the capacitor has little time between pulses to discharge; the average voltage across it therefore remains higher.

As the figure shows, the filter output voltage varies in amplitude with the frequency deviation. The original modulating signal is developed across the filter output. The filter components are carefully selected to minimize the ripple caused by the charging and discharging of the capacitor while at the same time providing the necessary high-frequency response for the original modulating signal. Some pulse-averaging discriminators generate a pulse every half-cycle or at every zero crossing instead of every one cycle of the input. With a greater number of pulses to average, the output signal is easier to filter and contains less ripple.

The pulse-averaging discriminator is a very high-quality frequency demodulator. In the past, its use was limited to expensive telemetry and industrial control applications. Today, with the availability of low-cost ICs, the pulse-averaging discriminator is easily implemented and is used in many electronic products.

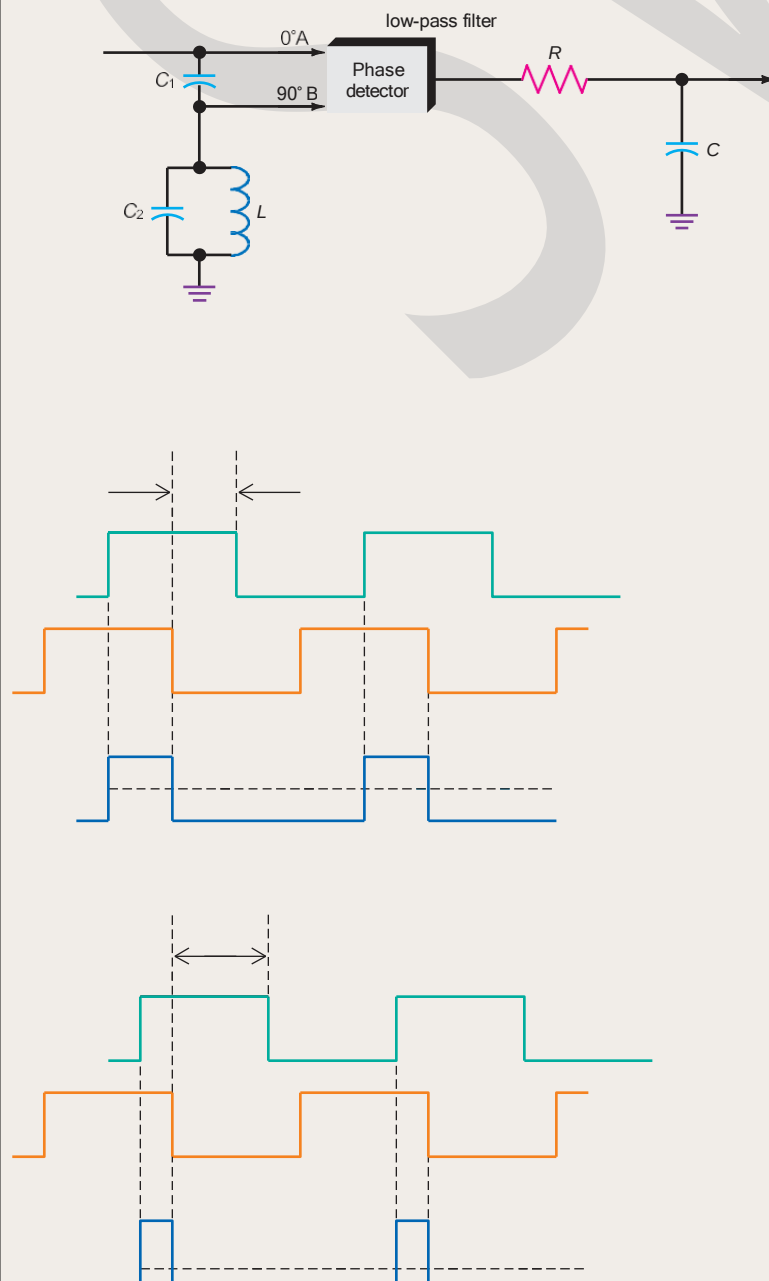
Quadrature Detectors

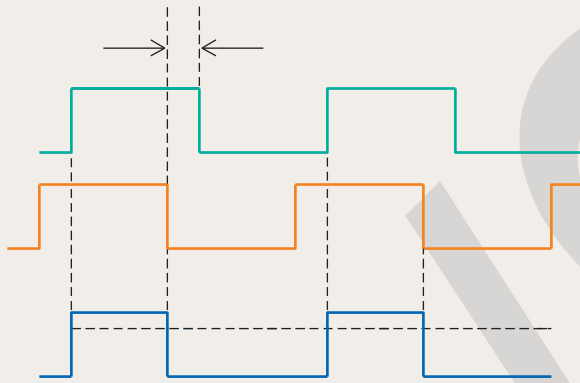
The *quadrature detector* uses a phase-shift circuit to produce a phase shift of 90° at the unmodulated carrier frequency. The most commonly used phase-shift arrangement is shown in Fig. 1-23. The frequency-modulated signal is applied through a very small capacitor (C_1) to the parallel-tuned circuit, which is adjusted to resonate at the center carrier frequency. At resonance, the tuned circuit appears as a high value of pure resistance. The small capacitor has a very high reactance compared to the tuned circuit impedance. Thus, the output across the tuned circuit at the carrier frequency is very close to 90° and leads the input. When frequency modulation occurs, the carrier frequency deviates above and below the resonant frequency of the tuned circuit, resulting in an increasing or a decreasing amount of phase shift between the input and the output.

The two quadrature signals are then fed to a phase detector circuit. The output of the phase detector is a series of pulses whose width varies with the amount of phase shift between the two signals. These signals are averaged in an RC low-pass filter to recreate the original modulating signal.

Normally the sinusoidal FM input signals to the phase detector are at a very high level and drive the differential amplifiers in the phase detector into cutoff and saturation. The differential transistors act as switches, so the output is a series of pulses. No limiter is needed if the input signal is large enough. The duration of the output pulse is determined by the amount of phase shift. The phase detector can be regarded as an AND gate whose output is ON only when the two input pulses are ON and is OFF if either one or both of the inputs are OFF.

Fig. 1-24 shows the typical waveforms involved in a quadrature detector. When there is no modulation, the two input signals are exactly 90° out of phase and therefore provide an output pulse with the indicated width. When the FM signal frequency increases, the amount of phase shift decreases, resulting in a wider output pulse. The wider pulses averaged by the RC filter produce a higher average output voltage, which corresponds to the higher amplitude required to produce the higher carrier frequency. When the signal frequency decreases, greater phase shift and narrower output pulses occur. The narrower pulses, when averaged, produce a lower average output voltage, which corresponds to the original lower-amplitude modulating signal.





Phase-Locked Loops

A *phase-locked loop* (PLL) is a frequency- or phase-sensitive feedback control circuit used in frequency demodulation, frequency synthesizers, and various filtering and signal detection applications. All phase-locked loops have the three basic elements, shown in Fig. 1-25.

1. A phase detector is used to compare the FM input, sometimes referred to as the *reference signal*, to the output of a VCO.
2. The VCO frequency is varied by the dc output voltage from a low-pass filter.
3. The low-pass filter smooths the output of the phase detector into a control voltage that varies the frequency of the VCO.

The primary job of the phase detector is to compare the two input signals and generate an output signal that, when filtered, will control the VCO. If there is a phase or frequency difference between the FM input and VCO signals, the phase detector output varies in proportion to the difference. The filtered output adjusts the VCO frequency in an attempt to correct for the original frequency or phase difference. This dc control voltage, called the *error signal*, is also the feedback in this circuit.

When no input signal is applied, the phase detector and low-pass filter outputs are zero. The VCO then operates at what is called the *free-running frequency*, its normal operating frequency as determined by internal frequency-determining components. When an input signal close to the frequency of the VCO is applied, the phase detector compares the VCO free-running frequency to the input frequency and produces an output voltage proportional to the frequency difference. Most PLL phase detectors operate just as the one discussed in the section on quadrature detectors.

The phase detector output is a series of pulses that vary in width in accordance with the amount of phase shift or frequency difference that exists between the two inputs. The output pulses are then filtered into a dc voltage that is applied to the VCO. This dc voltage is such that it forces the VCO frequency to move in a direction that reduces the dc error voltage. The error voltage forces the VCO frequency to change in the direction that reduces the amount of phase or frequency difference between the VCO and the input. At some point, the error voltage causes the VCO frequency to equal the input frequency; when this happens, the PLL is said to be in a *locked* condition. Although the input and VCO frequencies are equal, there is a phase difference between them, usually exactly 90° , which produces the dc output voltage that will cause the VCO to produce the frequency that keeps the circuit locked.

If the input frequency changes, the phase detector and low-pass filter produce a new value of dc control voltage that forces the VCO output frequency to change until it is equal to the new input frequency. Any variation in input frequency is matched by a VCO frequency change, so the circuit remains locked. The VCO in a PLL is, therefore, capable of *tracking* the input frequency over a wide range. The range of frequencies over which a PLL can track an input signal and remain locked is known as the *lock range*. The lock range is usually a band of frequencies above and below the free-running frequency of the VCO. If the input signal frequency is out of the lock

range, the PLL will not lock. When this occurs, the VCO output frequency jumps to its free-running frequency.

Superheterodyne Receivers

Superheterodyne receivers convert all incoming signals to a lower frequency, known as the *intermediate frequency (IF)*, at which a single set of amplifiers and filters is used to provide a fixed level of sensitivity and selectivity. Most of the gain and selectivity in a superheterodyne receiver are obtained in the IF amplifiers. The key circuit is the mixer, which acts as a simple amplitude modulator to produce sum and difference frequencies. The incoming signal is mixed with a local oscillator signal to produce this conversion.

Fig. 9-5 shows a general block diagram of a superheterodyne receiver

RF Amplifiers

The antenna picks up the weak radio signal and feeds it to the *RF amplifier*, also called a *low-noise amplifier (LNA)*. Because RF amplifiers provide some initial gain and selectivity, they are sometimes referred to as *preselectors*. Tuned circuits help select the desired signal or at least the frequency range in which the signal resides.

In communication receivers that do not use an RF amplifier, the antenna is connected directly to a tuned circuit, at the input to the mixer, which provides the desired initial selectivity. This configuration is practical in low-frequency applications where extra gain is simply not needed. (Most of the receiver gain is in the IF amplifier section, and even if relatively strong signals are to be received, additional RF gain is not necessary.)

RF amplifiers also minimize oscillator radiation. The local oscillator signal is relatively strong, and some of it can leak through and appear at the input to the mixer. If the mixer input is connected directly to the antenna, some of the local oscillator signal radiates, possibly causing interference to other nearby receivers. The RF amplifier between the mixer and the antenna isolates the two, significantly reducing any local oscillator radiation.

Mixers and Local Oscillators

The output of the RF amplifier is applied to the input of the *mixer*. The mixer also receives an input from a local oscillator or frequency synthesizer. The mixer output is the input signal, the local oscillator signal, and the sum and difference frequencies of these signals. Usually a tuned circuit at the output of the mixer selects the difference frequency, or intermediate frequency (IF). The sum frequency may also be selected as the IF in some applications. The mixer may be a diode, a balanced modulator, or a transistor. MOSFETs and hot carrier diodes are preferred as mixers because of their low-noise characteristics.

The *local oscillator* is made tunable so that its frequency can be adjusted over a relatively wide range. As the local-oscillator frequency is changed, the mixer translates a wide range of input frequencies to the fixed IF.

IF Amplifiers

The output of the mixer is an IF signal containing the same modulation that appeared on the input RF signal. This signal is amplified by one or more IF amplifier stages, and most of the receiver gain is obtained in these stages. Selective tuned circuits provide fixed selectivity. The intermediate frequency is usually much lower than the input signal frequency, IF amplifiers are easier to design and good selectivity is easier to obtain. Crystal, ceramic, or SAW filters are used in most IF sections to obtain good selectivity. Some forms of receivers use DSP filters for selectivity.

Demodulators

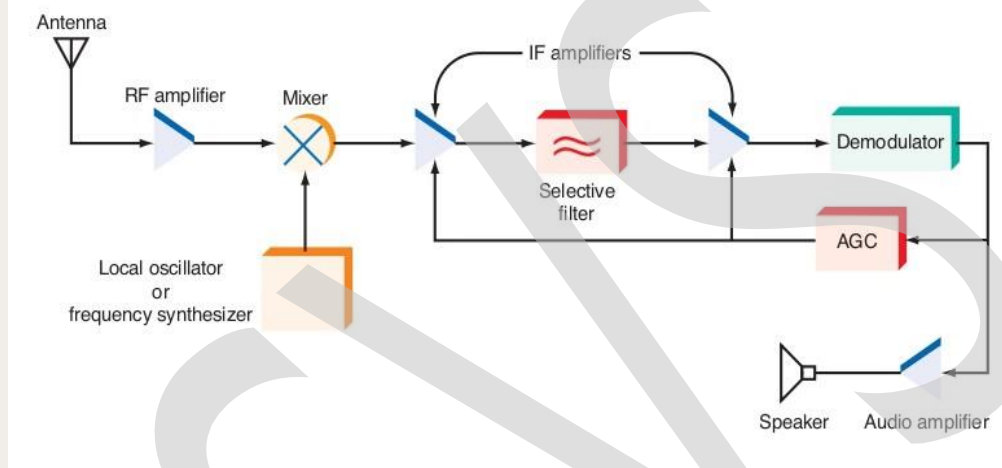
The highly amplified IF signal is finally applied to the *demodulator*, or *detector*, which recovers the original modulating information. In modern digital superheterodyne radios, the IF signal is first digitized by an analog-to-digital converter (ADC) and then sent to a digital signal processor (DSP) where the demodulation is carried out by a programmed algorithm. The recovered signal in digital form is then converted back to analog by a digital-to-analog converter (DAC). The output of the demodulator or DAC is then usually fed to an audio amplifier with sufficient voltage and power gain to operate a speaker. For nonvoice signals, the detector output may be sent elsewhere, to a TV, tablet, cell phone screen, computer, or some other device.

Automatic Gain Control

The output of a demodulator is usually the original modulating signal, the amplitude of which is directly proportional to the amplitude of the received signal. The recovered signal, which is usually ac, is rectified and filtered into a dc voltage by a circuit known as the *automatic gain control (AGC)* circuit. This dc voltage is fed back to the IF amplifiers, and sometimes the RF amplifier, to control receiver gain.

AGC circuits help maintain a constant output voltage level over a wide range of RF input signal levels; they also help the receiver to function over a wide range so that strong signals do not produce performance-degrading distortion. The amplitude of the RF signal at the antenna of a receiver can range from a fraction of a microvolt to thousands of microvolts; this wide signal range is known as the *dynamic range*. Typically, receivers are designed with very high gain so that weak signals can be reliably received. However, applying a very high-amplitude signal to a receiver causes the circuits to be overdriven, producing distortion and reducing intelligibility.

With AGC, the overall gain of the receiver is automatically adjusted depending on the input signal level. The signal amplitude at the output of the detector is proportional to the amplitude of the input signal; if it is very high, the AGC circuit produces a high dc output voltage, thereby reducing the gain of the IF amplifiers. This reduction in gain eliminates the distortion normally produced by a high-voltage input signal. When the incoming signal is weak, the detector output is low. The output of the AGC is then a smaller dc voltage. This causes the gain of the IF amplifiers to remain high, providing maximum amplification.

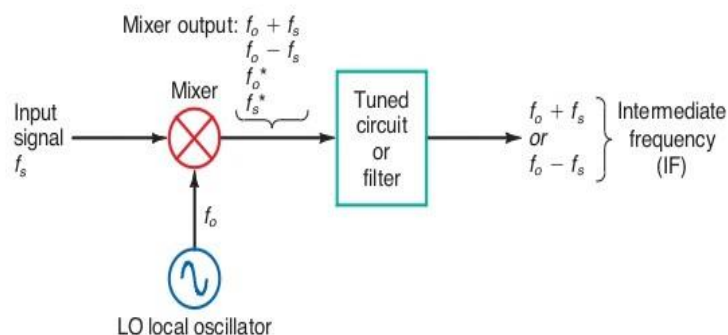
Figure 9-5 Block diagram of a superheterodyne receiver.

Frequency Conversion

Frequency conversion is the process of translating a modulated signal to a higher or lower frequency while retaining all the originally transmitted information. In radio receivers, high-frequency radio signals are regularly converted to a lower, intermediate frequency, where improved gain and selectivity can be obtained. This is called *down conversion*. In satellite communications, the original signal is generated at a lower frequency and then converted to a higher frequency for transmission. This is called *up conversion*.

Mixing Principles

Frequency conversion is a form of amplitude modulation or analog multiplication carried out by a mixer circuit or converter. The function performed by the mixer is called *heterodyning*.

Figure 9-6 Concept of a mixer.

* May or may not be in the output depending upon the type of mixer.

Fig. 9-6 is a schematic diagram of a mixer circuit. Mixers accept two inputs. The signal f_s , which is to be translated to another frequency, is applied to one input, and the sine wave from a local oscillator f_o is applied to the other input. The signal to be translated can be a simple sine wave or any complex modulated signal containing sidebands. Like an amplitude modulator, a mixer essentially performs a mathematical multiplication of its two input signals

The oscillator is the carrier, and the signal to be translated is the modulating signal. The output contains not only the carrier signal but also sidebands formed when the local oscillator and input signal are mixed. The output of the mixer, therefore, consists of signals f_s , f_o , $f_o + f_s$, and $f_o - f_s$

The local oscillator signal f_o usually appears in the mixer output, as does the original input signal f_s in some types of mixer circuits. These are not needed in the output and are therefore filtered out. Either the sum or difference frequency in the output is the desired signal. For example, to translate the input signal to a lower frequency, the lower sideband or difference signal $f_o - f_s$ is chosen. The local oscillator frequency will be chosen such that when the information signal is subtracted from it, a signal with the desired lower frequency is obtained. When translating to a higher frequency, the upper sideband or sum signal $f_o + f_s$ is chosen. Again, the local oscillator frequency determines what the new higher frequency will be. A tuned circuit or filter is used at the output of the mixer to select the desired signal and reject all the others.

Mixer and Converter Circuits

Diode Mixers. The primary characteristic of mixer circuits is nonlinearity. Any device or circuit whose output does not vary linearly with the input can be used as a mixer. For example, one of the most widely used types of mixer is the simple but effective diode modulator

A diode mixer circuit using a single diode is shown in Fig. 9-7. The input signal, which comes from an RF amplifier or, in some receivers, directly from the antenna, is applied to the primary winding of transformer T_1 . The signal is coupled to the secondary winding and applied to the diode mixer, and the

local oscillator signal is coupled to the diode by way of capacitor C_1 . The input and local oscillator signals are linearly added in this way and applied to the diode, which performs its nonlinear magic to produce the sum and difference frequencies. The output signals, including both inputs, are developed across the tuned circuit, which acts as a bandpass filter, selecting either the sum or difference frequency and eliminating the others.

Doubly Balanced Mixer. Balanced modulators are also widely used as mixers. These circuits eliminate the carrier from the output, making the job of filtering much easier. Any of the balanced modulators described previously can be used in mixing applications. Both the diode lattice balanced modulator and the integrated differential amplifier-type balanced modulator are quite effective in mixing applications. A version of the diode balanced modulator shown in Fig. 4-24, known as a *doubly balanced mixer* and illustrated in Fig. 9-8, is probably the single best mixer available, especially for VHF, UHF, and microwave frequencies. The transformers are precision-wound and the diodes matched in characteristics so that a high degree of carrier or local oscillator suppression occurs. In commercial products, the local oscillator attenuation is 50 to 60 dB or more.

FET Mixers. FETs make good mixers because they provide gain, have low noise, and offer a nearly perfect square-law response. An example is shown in Fig. 9-9. The *FET mixer* is biased so that it operates in the nonlinear portion of its range. The input signal is applied to the gate, and the local oscillator signal is coupled to the source. Again, the tuned circuit in the drain selects the difference frequency.

Figure 9-7 A simple diode mixer.

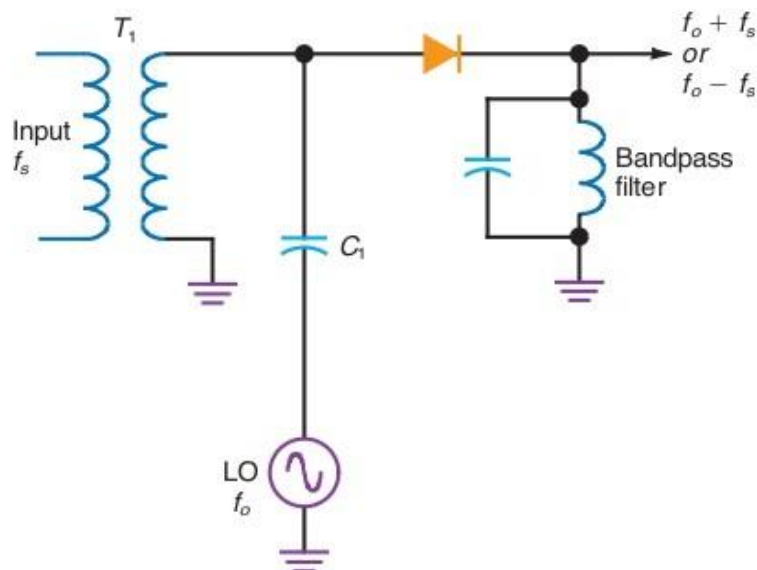


Figure 9-8 A doubly balanced mixer very popular at high frequencies.

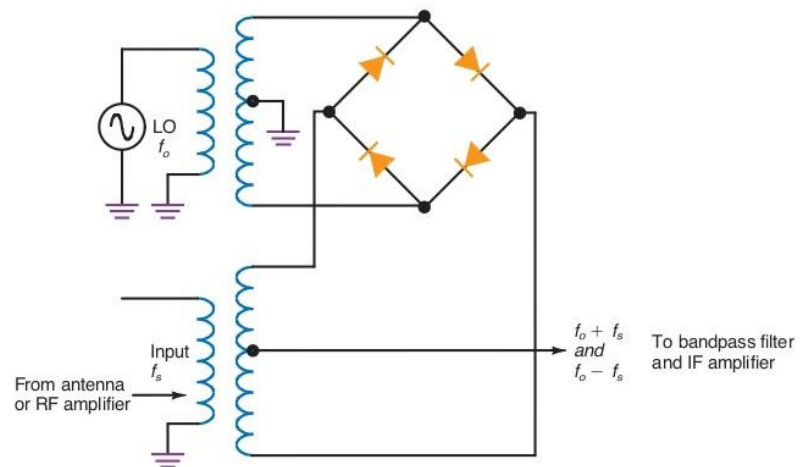


Figure 9-9 A JFET mixer.

